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$$s_4 = x_1 \cdot (1 + q + q^2 + q^3) = -20$$

$$s_3 - s_7 = -x_1(q^3 + q^4 + q^5 + q^6) = -x_1 \cdot q^3 \cdot (1 + q + q^2 + q^3)$$

$$\Rightarrow \frac{s_3 - s_7}{s_4} = \frac{-x_1 \cdot q^3 \cdot (1 + q + q^2 + q^3)}{x_1 \cdot (1 + q + q^2 + q^3)}$$

$$= \frac{540}{20} = 27$$

$$\Rightarrow -q^3 = 27 \Rightarrow q = -3$$

$$\Rightarrow x_1 = 1$$

References